Advanced Topics in Numerics of PDEs: FEM for the Navier–Stokes Equations (SS2026)

Chapter 2: Finite Element Computations

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April 15, 2026

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9. Adaptive mesh refinement
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11. Outlook & conclusions

Part 1:

Example: 1D Poisson problem

Strong form

Given Ω,

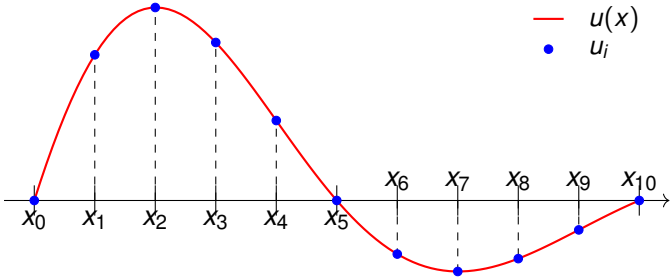
with boundary conditions

- Dirichlet boundary condition
- Neumann boundary condition

where $\partial\Omega = \Gamma_D \cup \Gamma_N$ and $\Gamma_D \cap \Gamma_N = \emptyset$.

Numerical solution approaches

Finite difference method (FDM):



Idea: Search approx. solution at points

$$u_i \approx u(x_i)$$

and approx. derivatives by FD.

Pros:

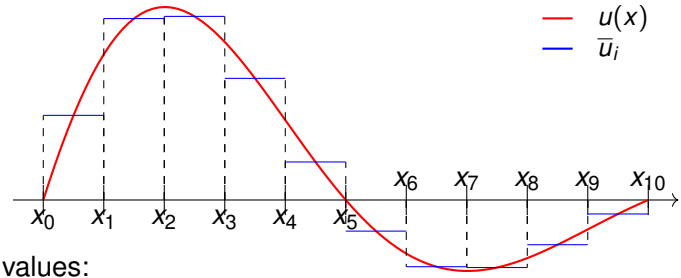
-

Cons:

-

Numerical solution approaches (cont.)

Finite volume method (FVM):



Idea: Search approx. cell-averaged values:

$$\overline{u}_i \approx \int_{x_i}^{x_{i+1}} u(x)dx \Big/ \int_{x_i}^{x_{i+1}} 1\,dx$$

Building blocks: reconstruction and fluxes at edges

Pros:

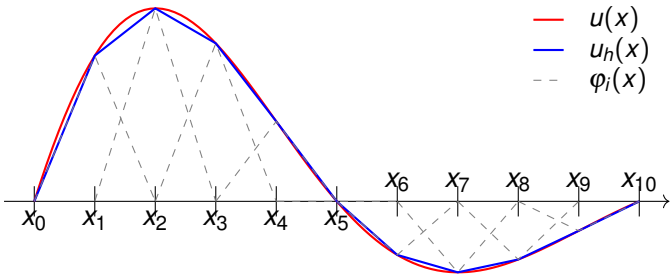
-
-

Cons:

-

Numerical solution approaches (cont.)

Galerkin framework:



Idea:

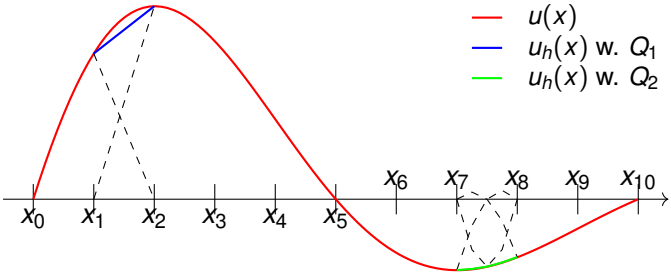
that minimizes the error $u_h - u(x)$. For this purpose, we work on the weak form.

Examples (based on the shape functions φ_i):

- spectral functions $\rightarrow$ spectral method
- hierarchical functions $\rightarrow$ hierarchical method
- piecewise functions defined on elements $\rightarrow$ finite element method
- ...

Numerical solution approaches (cont.)

Finite element method (FEM):



Piecewise functions, e.g., Lagrange function.

Pros:

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-
-

Cons:

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-
-

Variational formulation

The variational problem reads:

where

and the bilinear form $a(\cdot,\cdot)$ and linear functional $\ell(\cdot)$ are given by

Variational formulation: derivation

Step 1: Multiply by a test function
Let $v \in H^1(\Omega)$ with $v = 0$ on Γ_D. Multiply the equation by v and integrate over Ω:

$$\int_\Omega (-\Delta u)\, v\, dx = \int_\Omega f v\, dx.$$

Step 2: Integration by parts: Using Green's first identity:

$$\int_\Omega (-\Delta u)\, v\, dx = \int_\Omega \nabla u \cdot \nabla v\, dx - \int_{\partial\Omega} \frac{\partial u}{\partial n} v\, ds.$$

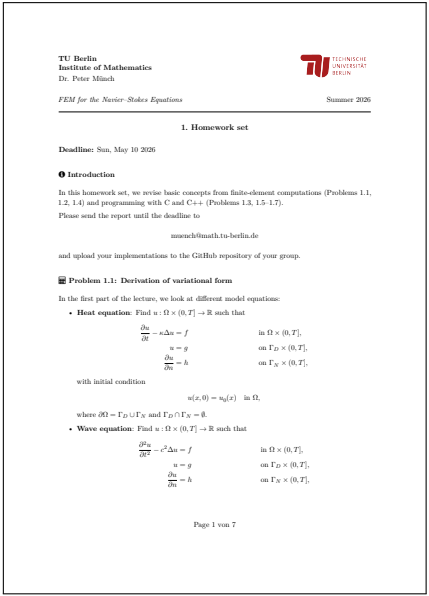
Step 3: Apply boundary conditions
Split the boundary integral. Since $v = 0$ on Γ_D, the Dirichlet term vanishes. Using the Neumann condition on Γ_N, we obtain:

$$\int_{\partial\Omega} \frac{\partial u}{\partial n} v\, ds = \int_{\Gamma_N} h v\, ds.$$

Step 4: Weak form: Thus, the weak formulation is:

$$\int_\Omega \nabla u \cdot \nabla v\, dx = \int_\Omega f v\, dx + \int_{\Gamma_N} h v\, ds.$$

Homework set 1



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FEM for the Navier–Stokes Equations Summer 2026

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Deadline: Sun, May 10 2026

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Problem 1.1: Derivation of variational form

In the first part of the lecture, we look at different model equations:

- **Heat equation**: Find $u : \Omega \times (0,T] \to \mathbb{R}$ such that

$$\begin{aligned} \frac{\partial u}{\partial t} - \kappa \Delta u &= f && \text{in } \Omega \times (0,T], \\ u &= g && \text{on } \Gamma_D \times (0,T], \\ \frac{\partial u}{\partial n} &= h && \text{on } \Gamma_N \times (0,T], \end{aligned}$$

  with initial condition

$$u(x,0) = u_0(x) \quad \text{in } \Omega,$$

  where $\partial\Omega = \Gamma_D \cup \Gamma_N$ and $\Gamma_D \cap \Gamma_N = \emptyset$.

- **Wave equation**: Find $u : \Omega \times (0,T] \to \mathbb{R}$ such that

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- Task 1: Derive variational form for other model problems

Discretized weak form

Let $V_h \subset V$ and $V_{0,h} \subset V_0$ be finite-dimensional subspaces. Find $u_h \in V_h$ such that

$$a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in V_{0,h}.$$

We set ($\varphi_j(x)$: shape functions, U_j: unknown coefficients)

$$u_h(x) = \sum_{j=1}^{N} U_j \varphi_j(x), \quad v_h(x) = \sum_{j=1}^{N} V_j \varphi_j(x).$$

.

Algebraic system

We get the algebraic system:

with the stiffness matrix K and load vector F given by

Algebraic system (cont.)

Example: 1D domain [0, 1] with homogeneous DBC at left and right end and equidistant mesh with $\Delta x = 1/(N+1)$

Stiffness matrix $K \in R^{N \times N}$:

<u>Properties:</u>

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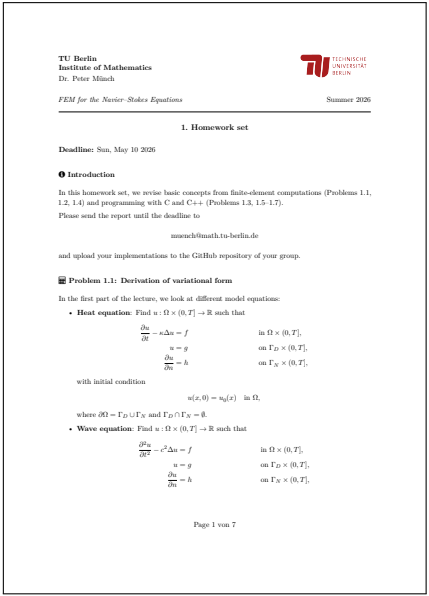
-
-

Homework set 1



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- Task 2: compute mass matrix

Implementation details

In the context of FEM, we replace the integration

$$a(\varphi_i, \varphi_j) = \int_\Omega \nabla \varphi_i \cdot \nabla \varphi_j \, dx$$

by a

Implementation details (cont.)

We process all shape function that have a support on a cell together, resulting in *element matrices* K_c (partial results) that need to be *assembled* into the global stiffness matrix K.

Notation:

$$K = \sum_c R_c^\top K_c R_c,$$

where the restriction matrix R_c is mapping local indices (of shape functions) to global ones.

Part 2:

Error estimates

Interpolation error

Definition: Polynomial interpolation is given by polynomial of degree n ($p_n : [a, b] \to \mathbb{R}$):

$$p_n(x) = a_0 + a_1 x + \ldots + a_n x^n$$

such that the interpolation condition is fulfilled

$$p_n(x_i) = y_i, \quad i = 0, \ldots, n.$$

Consider that y_i is given by the function $f : [a, b] \to \mathbb{R}$:

$$y_i = f(x_i), \; i = 0, \ldots, n.$$

Definition: interpolation error at distinct data points

$$e_n(x) = f(x) - p_n(x) \quad \text{on} \quad x \in [a, b].$$

Interpolation error (cont.)

Theorem: Let $f \in C^{n+1}$ on $[a,b]$. Then, there is, for each $x \in [a,b]$, a $\xi \in [a,b]$ such that:

$$\boxed{e_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} w_{n+1}(x)} \quad \text{mit} \quad w_{n+1}(x) = (x - x_0)(x - x_1) \cdots (x - x_n).$$

This implies

$$\text{and} \quad ||\nabla e||_{L^2} \leq \frac{C(p)}{p!} h^p.$$

More details: Numerical mathematics I

A priori error estimate

Interpolation error and Galerkin orthogonality

$$(\nabla\varphi_h, \nabla(u - u_h)) = 0, \quad \forall\varphi_h \in V_h \subset H^1$$

give, for the Poisson problem, the <u>a-priori error estimates</u>:

$$\text{and} \quad ||\nabla(u - u_h)||_{L^2} < Ch^p ||\nabla^{p+1} u||_{L^2}.$$

More details: Numerical mathematics III

A posteriori error estimate

Motivation: given u_h, determine the error, e.g., in the energy norm

$$||e_h||_A^2 = A(e_h, e_h) = ?$$

Long derivation gives the <u>a posteriori error estimate</u>

$$\boxed{||e_h||_A \leq C \sqrt{\sum_T h_T^2 ||R_T||_T^2 + h_T ||J_E||_{\partial T}^2}}$$

with

- $R_K = f + \Delta u_h$
  cell residual: how well PDE is satisfied?
- $J_E = -\frac{1}{2}[[\partial_n u_h]]$
  edge residual: how smooth is the solution?

A posteriori error estimate (cont.)

Algorithm of adaptive mesh refinement (AMR):

$$\text{solve} \rightarrow \text{mark} \rightarrow \text{refine} \rightarrow \text{repeat}$$

Option: mark cells with high (a posteriori) error

$$h_T^2 ||R_T||_T^2 + h_T ||J_E||_{\partial T}^2.$$

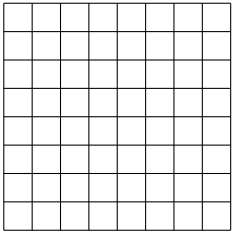
More details: Chapter 9 (AMR)

Part 3:

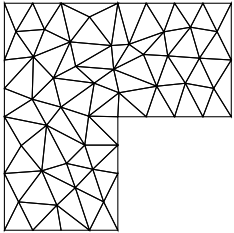
Extension to higher dimensions

Extension to 2D

structured mesh



unstructured mesh



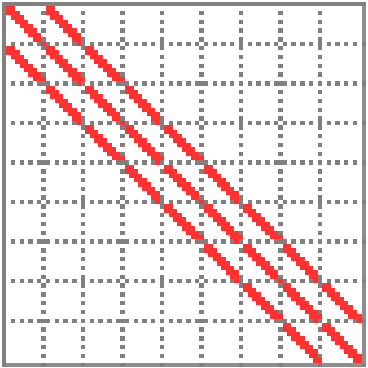
(Discrete) variational form still holds!

We only need to redefine the shape functions, e.g., linear shape functions (on triangles):

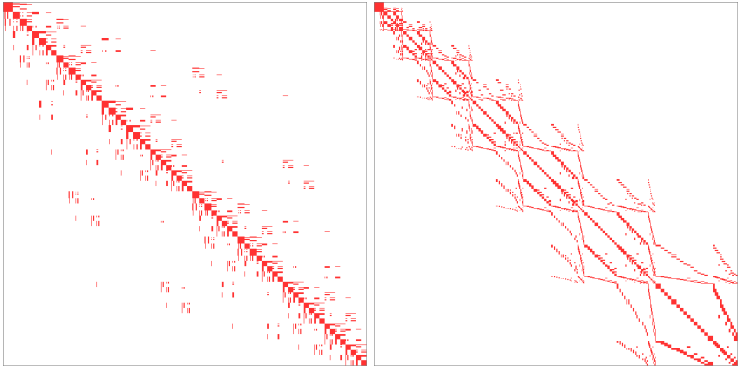
or bilinear shape functions (on quadrilaterals):

the quadrature, and the mapping (not shown; discussed later).

Sparsity patterns



structured mesh

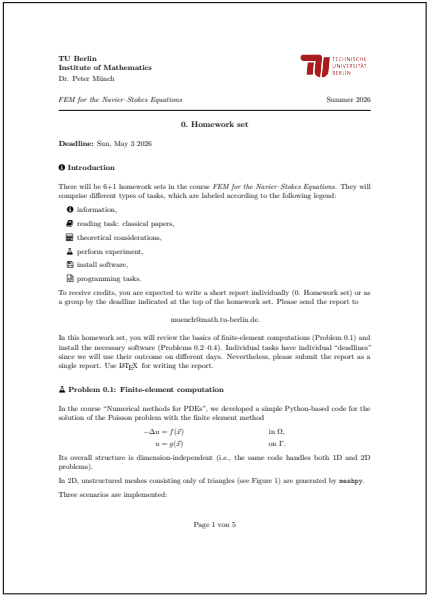


unstructured mesh

Part 4:

Implementation

Homework set 0



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Problem 0.1: Finite-element computation

In the course "Numerical methods for PDEs", we developed a simple Python-based code for the solution of the Poisson problem with the finite element method

$$-\Delta u = f(\vec{x}) \quad \text{in } \Omega,$$
$$u = g(\vec{x}) \quad \text{on } \Gamma.$$

Its overall structure is dimension-independent (i.e., the same code handles both 1D and 2D problems).

In 2D, unstructured meshes consisting only of triangles (see Figure 1) are generated by `meshpy`.

Three scenarios are implemented:

Page 1 von 5

- Task 1: review FEM code
- Task 2-4: install required software

Implementation

In the course “Numerical methods for PDEs”, we developed a simple FEM implementation. Scope:

- unstructured meshes,
- 1D (lines) and 2D (triangles),
- linear Lagrange elements for 1D and 2D, quadratic Lagrange elements for 1D,
- Gauss quadrature,
- isoparametric mapping.

Let’s discuss the implementation!

The following slides only contain the assembly of the system matrix and the right-hand-side vector.

Implementation: assembly

⓪
①
②

```
for cell, element in zip(mesh.elements, dofs.elements):

    n_dofs_per_cell = len(element)

    # compute quadrature points, Jacobian and its determinant
    ps = [mesh.points[i] for i in cell]
    J = np.transpose(np.linalg.inv(compute_Jacobian(dim, ps)))
    detJ = compute_Jacobian_determinant(dim, ps)

    # evaluate the value and gradient of the
    # shape functions at the quadrature points
    shape_value = np.zeros([n_dofs_per_cell, n_q_points])
    shape_gradient = np.zeros([n_dofs_per_cell, n_q_points, dim])

    for q, [x_ref, w] in enumerate(quadrature):
        shape_value[:, q] = reference_shape_value(dim, fe_degree, x_ref)
        shape_gradient[:, q, :] = np.transpose(
            J @ reference_shape_gradient(dim, fe_degree, x_ref))

    # see next slide
```

Implementation: assembly (cont.)

② loop over all cells: ⓑ

```
for cell, element in zip(mesh.elements, dofs.elements):

    # see previous slide

    A_cell = np.zeros([n_dofs_per_cell, n_dofs_per_cell])
    rhs_cell = np.zeros([n_dofs_per_cell])

    for q, [x_ref, w] in enumerate(quadrature):
        value = fu_f(compute_quadrature_point(dim, ps, x_ref))

        for i in range(0, n_dofs_per_cell):
            rhs_cell[i] += shape_value[i, q] * value * detJ * w

        for i in range(0, n_dofs_per_cell):
            for j in range(0, n_dofs_per_cell):
                A_cell[i, j] += np.dot(shape_gradient[i, q, :],
                                       shape_gradient[j, q, :]) * detJ * w

    # see next slide
```

Implementation: assembly (cont.)

② loop over all cells: ⓒ

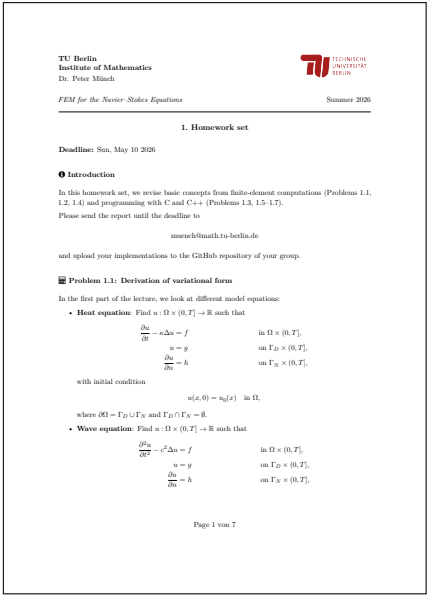
```
for cell, element in zip(mesh.elements, dofs.elements):
    # see previous slide

    # apply boundary condition
    for e, i in enumerate(element):
        if i in boundary_dofs:
            rhs_cell[e] = 0

            for j in range(0, len(element)):
                A_cell[e, j] = 0

                if not element[j] in boundary_dofs:
                    rhs_cell[j] -= A_cell[j, e] * fu_g(dofs.points[i])

                A_cell[j, e] = 0
            A_cell[e, e] = 1

    # assemble vector and matrix
    for e, i in enumerate(element):
        rhs_vector[i] += rhs_cell[e]

    for ej, j in enumerate(element):
        for ei, i in enumerate(element):
            A[i, j] += A_cell[ei, ej]
```

③

④

Workflow

Homework set 1



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- Task 3: extend Python code with quadrilaterals

Historical background: deal.II, for a long time, only supported hypercube cells (lines, quadrilaterals, hexahedra). Support for simplex and mixed meshes was added only recently (2021).

Part 5:

Conclusions & outlook

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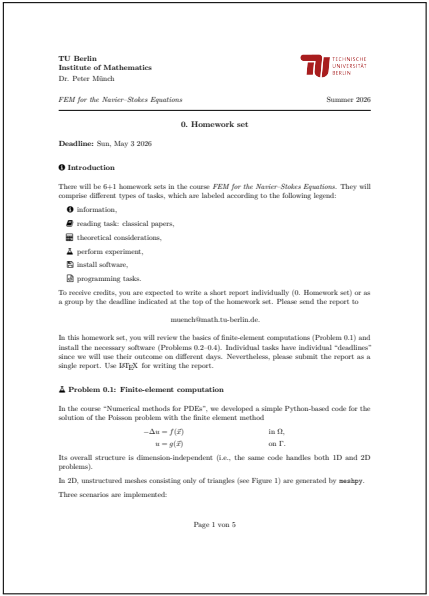

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- Task 2: install a terminal
- Task 3: install software, e.g., gcc
- Task 4: install deal.II